



DBMS PROJECT

25CS1302E

Smart Restaurant Ordering System

Project Abstract & Software Requirements

Database Systems Engineering and Distributed Backend
Development

Y25 · 2026-2027 · Trimester 4

DATABASE-DRIVEN

JAVA

MYSQL

JDBC

Project Abstract

A centralized, database-driven approach to restaurant ordering and management.

- The system is designed to improve the efficiency and convenience of restaurant ordering and management.
- It replaces manual order taking, which can lead to incorrect orders, delays, tracking difficulty, and higher staff workload.
- Customers can view the available menu, select food items, specify quantities, and place orders.
- Restaurant staff can manage menu items, update order status, and maintain customer, order, and billing information in a structured database.
- The system calculates the total bill from selected items and quantities while supporting organized and consistent data management.

Primary Goal

Reduce manual effort, minimize ordering errors, improve order processing, and keep restaurant data organized.

Data Managed

Customer • Menu • Order • Billing

Stored in related database tables for efficient retrieval and consistency.

Problem → Proposed Solution

Why the Smart Restaurant Ordering System is needed.

Challenges with Manual Ordering

- Incorrect orders can occur during manual order taking.
- Order processing and tracking may be delayed or difficult.
- Staff workload increases when menu, customer, billing, and order status records are handled manually.
- Maintaining accurate and organized records becomes harder.



Centralized Database Solution

- Customers access the available menu and place itemized orders.
- Restaurant staff manage menu items and update order statuses.
- Customer, menu, order, and billing data are maintained in structured tables.
- The system calculates the bill and supports accurate, efficient data retrieval.

Core Functional Scope & DBMS Concepts

Functions described in the abstract and the database concepts applied to support them.

Customer & Ordering

- View available menu
- Select food items
- Specify quantities
- Place orders

Restaurant Operations

- Manage menu items
- Update order status
- Track customer orders
- Calculate total bill

Database Foundation

- Tables & relationships
- Primary & foreign keys
- SQL queries & joins
- Constraints & transactions

Outcome Data consistency, efficient retrieval, reduced manual effort, fewer ordering errors, and more organized restaurant data management.

Software Requirements

Frameworks, platforms, and tools specified in the project abstract form.

Requirement Area	Technology / Tool
Front end	HTML, CSS, JavaScript
Backend	Java
Database	MySQL
Database Connectivity	JDBC
Development Environment	Visual Studio Code / IntelliJ IDEA
Database Tool	MySQL Workbench

HTML / CSS / JS

JAVA

MYSQL

JDBC

VS CODE / INTELLIJ

MYSQL WORKBENCH

Project Methodology

Proposed development approach for building the Smart Restaurant Ordering System.

1. Requirement & Planning

Understand customer and staff workflows
Define menu, customer, order & billing data
List required system functions
Plan the project modules

2. Database & Backend Design

Design related MySQL tables
Set primary & foreign keys
Apply constraints, queries & joins
Develop Java logic and JDBC connectivity

3. Interface, Testing & Integration

Build the HTML/CSS/JavaScript interface
Connect UI → Java → MySQL
Test ordering, billing & status updates
Check data consistency and error handling

Result

Integrated prototype flow: Menu → Order → Billing → Order Status, supported by a structured MySQL database.